Trigonometry & Additional Topics

Section 1: Trigonometric Ratios & Cofunctions

1 If $\sin(x^\circ) = \cos(42^\circ)$, where $0 < x < 90$, what is the value of x?

a. 42

b. 48

c. 52

d. 138

2 In right triangle ABC with right angle at C, if $\cos A = \frac{5}{13}$, what is the value of $\tan A$?

a. $\frac{5}{12}$

b. $\frac{12}{5}$

c. $\frac{12}{13}$

d. $\frac{13}{12}$

3 Which trigonometric ratio is equal to $\frac{1}{\cos\theta}$?

a. $\sin\theta$

b. $\cot\theta$

c. $\sec\theta$

d. $\csc\theta$

Section 2: Unit Circle & Radians

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4 Convert 210° to radians.

a. $\frac{5\pi}{6}$

b. $\frac{7\pi}{6}$

c. $\frac{4\pi}{3}$

d. $\frac{11\pi}{6}$

5 What is the exact value of $\sin(\frac{4\pi}{3})$?

a. $-\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{3}}{2}$

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

6 If θ is in Quadrant IV and $\cos\theta = \frac{3}{5}$, what is $\sin\theta$?

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

Section 3: Complex Numbers & Arithmetic

7 What is the value of $(3 + 4i)(2 - i)$ where $i^2 = -1$?

a. $10 + 5i$

b. $2 + 5i$

c. $6 - 4i$

d. $10 - 5i$

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8 What is the simplified form of $\frac{10}{3-i}$?

a. $3+i$

b. $3-i$

c. $\frac{3+i}{2}$

d. $3+2i$

9 What is the value of i^{102}?

a. 1

b. -1

c. i

d. $-i$

Section 4: Coordinate Geometry of Circles

10 What is the radius of the circle defined by $(x+4)^2+(y-7)^2=81$?

a. 7

b. 9

c. 18

d. 81

11 What is the center of the circle $x^2+y^2-10x+6y+9=0$?

a. $(5,-3)$

b. $(-5,3)$

c. $(10,-6)$

d. $(-10,6)$

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12 Which of the following points lies on the circle $(x-1)^2+(y-2)^2=25$?

a. $(4, 6)$

b. $(5, 5)$

c. $(1, 7)$

d. $(6, 2)$

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